

Owais Siddiqi

Bioinformatics Data Scientist | Computational Biology

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Professional Summary

Computational Biologist and Bioinformatics Data Scientist with three years of industry experience analysing large-scale genomic and transcriptomic datasets and building predictive models for patient stratification in oncology. Proven track record of translating multi-omics and clinical data into deployable diagnostic tools, from RNA-seq workflow design and ML model validation through to production ETL pipelines embedded in R&D workflows. Experienced collaborating across scientific, engineering, and clinical teams in fast-paced, purpose-led environments.

Technical Skills

- **Bioinformatics and Genomics:** Next-generation sequencing (NGS) pipeline development (FASTQ/BAM to VCF, RNA-seq to differential expression), QC, alignment, variant calling, fusion detection; STAR, STAR-Fusion, Salmon, Strelkatz, GATK, VEP, DESeq2, EdgeR, NetMHCpan.
- **Pipelines and Cloud:** Nextflow (DSL2), Docker, CI/CD, Google Cloud Platform, HPC, Linux; reproducible and containerised workflow design.
- **Programming and Data:** Python (production pipelines), R, SQL, Bash, Git; Pandas, NumPy, Jupyter.
- **Machine Learning and AI:** Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch. Classification, survival analysis (Cox PH, time-varying Cox, Kaplan-Meier), clustering (HDBSCAN, GMM); preparing ML-ready datasets.
- **Data Visualisation and Reporting:** Streamlit, Plotly, Matplotlib, Seaborn, Power BI, Excel.

Work Experience

Willows Health

Apr 2026 – Present

Data Scientist (seconded to CB1 Medical)

- **Data Engineering:** Engineered SQL workflows to extract, reconcile, and map structured and semi-structured data across CRM, prescribing, and operational systems, consolidating six disparate sources into a single source of truth for downstream analysis.
- **Forecasting and Automation:** Built forecasting models on live operational data, improving projection accuracy by approximately 25%, and deployed automation tooling that cut manual reporting effort by around 8 hours per week.

CellBxHealth plc

Jan 2025 – Mar 2026

Data Scientist

- **NGS Pipeline Development:** Designed and implemented a novel RNA-seq workflow for rare-cell (CTC) transcriptomics, leading experimental design, QC benchmarking, differential expression, and biological interpretation. Reduced per-cohort processing from 6 to 4 working days and secured £150k in internal funding to co-develop a commercial library-prep QC product with a genomics-sequencing partner.
- **Reproducible QC Framework:** Built a bespoke, per-stage QC framework from scratch for a library-prep method never previously applied to CTCs, validating linearity against spiked inputs and enforcing reproducibility across experimental and computational layers where no reference benchmarks existed.
- **Data Engineering and ETL:** Developed automated ETL pipelines integrating imaging, NGS, and clinical meta-data into structured, analysis-ready datasets, reducing onboarding and QC processing time by 33% per study.

- **ML for Patient Stratification:** Prepared ML-ready feature sets from merged imaging and molecular data and designed, trained, and validated supervised classification models for patient stratification, adopted into scientist workflows for downstream analysis.
- **User Enablement:** Introduced structured SOPs and documentation and presented analyses to R&D, commercial, and senior leadership, eliminating approximately 8 hours of manual processing time per week across the team.

Genevation Ltd.

Apr 2024 – Dec 2024

Junior Genomic Data Scientist

- **NGS Workflow Automation:** Engineered containerised FASTQ-to-VCF pipelines on GCP and HPC (STAR, Strelka2, GATK, VEP), reducing average analysis turnaround from 5 to 2 days per sample batch and delivering reproducible genomic outputs for internal R&D.
- **AI-Driven Biomarker Discovery:** Processed somatic variants through NetMHCpan for MHC-binding prediction and trained a cancer-specific XGBoost classifier on the outputs, integrated directly into the pipeline to prioritise neoantigen candidates per patient.
- **Fundraising Support:** Produced analytical outputs and visualisations incorporated into investor pitch decks that supported £400k in fundraising.

PDUK Ltd.

Aug 2023 – Jan 2024

Data Scientist Intern

- **Data Engineering:** Built SQL-based cleansing and transformation workflows across a large customer database, reducing batch processing time by 25%.
- **Modelling and Reporting:** Developed demand-forecasting and k-means segmentation models feeding leadership dashboards, informing commercial planning and inventory decisions.

Research Experience

- **Owais Siddiqi**, Winfield J., Zormpas-Petridis K., Harris E., et al. *Investigating the Repeatability of Multifrequency Magnetic Resonance Elastography Applied to a Soft Gelatin Phantom*. **31st ISMRM Conference & Exhibition**, London, UK, 2022.
- Ciccioli M., Spode M., Coates M., Duhig E., **Owais Siddiqi**, et al. *Unlocking the Potential of Liquid Biopsy in Glioblastoma: An Extensive Evaluation of Circulating Biomarkers*. **SNO Annual Meeting**, 2025.
- Mezni A., Goodwin C., Greaves D., **Owais Siddiqi**, et al. *The Value of Characterising Circulating Tumour Cells Using Epithelial and Mesenchymal Markers in Breast Cancer*. **7th ACTC Conference**, 2025.
- Geraldizo M.C., Coates M., Torkzadeh F., **Owais Siddiqi**, et al. *Custom Gene Expression Profiling of Circulating Tumour Cells via Amplitude-Based Multiplex Digital PCR*. **7th ACTC Conference**, 2025.

Education

Imperial College London

2022 – 2023

MSc (Honours) Biomedical Engineering, Merit

- **CNN Regression Modelling:** Designed and trained a TensorFlow/Keras CNN to predict cell differentiation from microscopy images for phenotypic screening, achieving an R^2 of 0.80. Applied 5-fold cross-validation with hyperparameter tuning, and regularised against overfitting using dropout, L1/L2 penalties, and data augmentation.

Queen Mary University of London

2019 – 2022

BEng (Honours) Biomedical Engineering, First Class

- **BCI Classification Project:** Built a multi-class EEG-SSVEP classifier in Python and MATLAB, implementing a bagging ensemble of ANFIS binary classifiers using OAO decomposition and average voting aggregation.